TURNOVER, GENDER AND SUSTAINABILITY: EVIDENCE FROM THE GLOBAL MICROFINANCE INDUSTRY

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Abstract

Although microfinance remains a preferred development tool in many developing countries in the world, the sector is now facing many challenges and struggling to maintain its sustainability in recent years. This study aims to examine the effects of turnover (employee and borrower) and gender on various sustainability dimensions of MFIs. Consequently, we have utilized a very recent and globally represented unbalanced panel data of 2,032 microfinance institutions (MFIs) from 2010 to 2018. After considering the linear and quadratic relationship existing between turnover and sustainability of MFIs through fixed effect modeling, we observed significant and robust results in the linear models, and an opposite in the quadratic models for most of the cases. Interestingly, gender exhibits a conflicting effect on the sustainability of MFIs. As an instance, female managers were observed to have a positive and statistically significant effect on financial sustainability and social outreach, while female loan officers were found to be at variance, posing a threat to the financial autonomy of MFIs. The outcome of this paper proffers some insights to concerned stakeholders which are subsequently discussed in the policy implications.

Key words: Microfinance; Financial sustainability; Social outreach; Cost sustainability; Survivability; Microfinance institutions.

1. Introduction

Recently, microfinance institutions (MFIs) have been experiencing numerous challenges which include attainment of financial viability without subsidies or donations (Allet & Hudon, 2015; Lam, Zhang, Ang, & Jacob, 2020; Manos & Yaron, 2009; Mia, Zhang, Zhang, & Kim, 2018), commercialization of operation or mission drift (Beisland, D'Espallier, & Mersland, 2019; De Quidt, Fetzer, & Ghatak, 2018; Mia & Lee, 2017), capital constraints (Gupta, 2017; Tchuigoua, Durrieu, & Kouao, 2017), and intense competition and pressure from peers and industry regulators respectively (Butcher & Galbraith, 2019; Djan & Mersland, 2017; Mia, 2018). However, little is known about the contribution of the employee and borrower turnovers to the existing problems faced by MFIs. Surprisingly, the turnover rates (both employee and borrower) in the microfinance industry are relatively high (above 20% except staff turnover for the year 2010 and 2011) in the global context (Figure 1). Motivated by the paucity of literature on this topic, this study empirically investigates the effect of employees'[1] and borrowers' turnover on various dimensions of MFIs sustainability[2] from an institutional perspective.

[1] Employee and staff were used interchangeably in this study.
[2] Despite the subtle difference between sustainability and performance, they were utilized interchangeably in this study for simplicity. A brief discussion on sustainability is presented in Section 2.1

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The human asset is considered a significant factor in organizations' key competency and competitive advantage over their peers (Campbell, Ganco, Franco, & Agarwal, 2012; Coff, 1997; Lippman & Rumelt, 1982). However, employee's mobility is a common phenomenon experienced by every organization and may vary in degree between sectors and levels of job. In terms of the effects of turnovers on firms' or organizations' performance, there are at least two conflicting thoughts. On one hand, employee's turnover is considered a loss of human and social capital which in turn results in operational disruptions; and hence, a negative effect on organizational performance based on the traditional beliefs in management literature (De Winne, Marescaux, Sels, Van Beveren, & Vanormelingen, 2019; Hancock, Allen, Bosco, McDaniel, & Pierce, 2013).

On the other hand, the relationship between employee's turnover and organization performance may present a new twist. For example, employee's turnover will up to an optimal level result in lower labor costs and invention of innovative ideas (De Winne et al., 2019). Subsequent to the attainment of the optimal threshold, the drawback of turnover outweighs the benefits; consequently resulting in a decline in the organization's performance (Shaw, 2011; Shaw, Gupta, & Delery, 2005). In this regard, the relationship is rather an inverted U-shaped, as opposed to being linear. Lee (2018) further analyzed the effects of different employee turnovers on organizational performance and concluded that employees' transfer and resignation exhibit an inverted U-shaped and U-shaped relationship respectively, with involuntary turnover portraying a linear and positive relationship with organization's performance. In summary, turnover will affect the performance of an organization; however, the existence of a linear or non-linear relationship between them remains inconclusive.

Similar with the mobility of employees, a borrower/client turnover may also occur either through voluntary dropout or expulsion of MFIs (Murphy, 2018). Some of the factors that could trigger the withdrawal of a borrower from an MFI include loan size, loan types, saving facilities, non-financial products, after service monitoring, and personal factors (Cohen, 2002; Escalante & Rusiana, 2018; Meyer, 2002; Nayak, 2014; Pearlman, 2014). Retaining microfinance clients is not only crucial for MFIs financial sustainability, but also for value generation through serving the poor; which remains the core business strategy of MFIs (Urquizo, 2006). Therefore, customers should be considered an important asset through which long-term sustainability of the firm can be attained (Gupta, Lehmann, & Stuart, 2004). Despite the fact that clients' dropout is inevitable, its effects on the sustainability of MFIs still remain largely unexplored, and therefore warrant proper investigation.

Although there is a surge in academic literature on various issues of microfinance in recent years, studies on the turnover effects of microfinance sustainability is very limited to the author's knowledge. One of the reasons for such limitation could be attributed to lack of access to data (particularly on turnover rates). However, access to such data has been made possible since 2019 by the joint efforts of the World Bank and Microfinance Information Exchange, thereby enabling us to conduct this study. Given the paucity of research in this aspect of microfinance coupled with the inconclusive relationship existent between turnover and MFIs' performance, we are poised to pioneer a comprehensive study to establish the existence of linear or non-linear relationship between turnover (employee and borrower) and the various dimensions of MFIs sustainability.

Since sustainability is a complex concept, we have focused on various proxies of sustainability parameters such as cost sustainability, social outreach goal, financial sustainability, and institutional survivability. The findings will also offer the managers and practitioners valuable insights on the effects of turnover on organizational performance, which will in turn motivate them to draft policies to mitigate its consequences. Similarly, this study is unique as it also investigates the effects of borrowers' turnover; thus, offering a robust observation which is lacking in the existing literature. Nonetheless, the inclusion of gender variables in this study will also revisit the contribution of gender to MFIs' sustainability. Besides, modeling of turnover, gender and sustainability dimensions of MFIs will lay the foundation for further research related to the topic. By testing the linear and non-linear relationship of turnover on MFIs' sustainability, the study will contribute to the recent debate on human resource literature. Since we employed recent global data spanning from 2010 to 2018, the findings of this study can be enunciated for the global microfinance industry rather than a specific region or country.

The rest of the study is organized as follows: section 2 briefly reviews some of the existing theories on turnover and firms' performance; section 3 discusses the methodology; section 4 presents the results and discussion; and lastly, section 5 covers the conclusion, policy implications, and future research directions.

2. Brief Literature Review

In this section, we briefly discussed about sustainability and microfinance; and subsequently, a brief overview of turnover and its consequences on organizational performance. For brevity of the literature review, we have presented a lengthy discussion on existing literature (empirical and conceptual) about gender and performance of MFIs in the modeling section under the methodology.

2.1 Sustainability and Microfinance

Sustainability in microfinance is undoubtedly important; (Karmakar, 2008) and without it, MFIs could cause more harm than good (Bhanot & Bapat, 2015). The claim has also been substantiated by Schreiner (2000) who profoundly argued that "unsustainable MFIs might help the poor now, but they will not help the poor in future because the MFIs will be gone (p. 425)." Surveying the microfinance literature on sustainability, one of the earliest works accredited to Yaron, Benjamin, and Piprek (1997), emphasized the role of institutional outreach and financial viability on MFIs. Subsequently, the two roles have become the main focus of practitioners and policymakers. To reiterate, Hulme (2000) argued that once these two goals have been achieved, the sustainability of the microfinance industry is deemed beneficial.

The economic aspects of sustainability also refer to financial sustainability or economic viability, which denote MFIs' ability to maintain their operations over time (Schreiner, 2000). Britzelmaier, Kraus, and Xu (2013) defined financial sustainability as the ability of MFIs to generate enough revenue to cover total operational cost, in order to survive and prosper in the long-run. The social aspect of sustainability in microfinance refer to how well MFIs uphold the access of the poor to financial services, in an ongoing effort to alleviate poverty and empower women in patriarchal societies (Kabeer, 2005; Mosley, 2001; Mosley & Rock, 2004). Toindepi (2016) and Hudon (2009) also contended that sustainability would not be achieved until MFIs meet the financial needs of the poor. Although Gibson (2012) has highlighted eight dimensions of sustainability of MFIs, we will only be covering four dimensions (institutional, mission, human resources, and financial sustainability) in this study, given the available data.

2.2 Turnover and Performance of an Organization

Given the complex relationship between employees' turnover and organizational performance, researchers have resorted to many theories/perspectives to better understand the dynamics existent between the two. In these circumstances, Lee (2018); De Winne et al. (2019) and Hancock et al. (2013) have expounded various theories, and provided different scenarios of employees' turnover effect on organizational performance. We will briefly discuss some of these theories in light of the aforementioned studies.

One of the most dominant perspective in this aspect is the dysfunctional effect of employees' turnover on performance of an organization, resulting in a negative and linear relationship. Hence, any increase in turnover rate will affect an organization (Lee, 2018; Park & Shaw, 2013). The negative and linear effects of employees' turnover can be explained by three factors as highlighted by Lee (2018); namely, monetary cost (Allen, Bryant, & Vardaman, 2010; Dalton & Todor, 1979; Staw, 1980), loss of accumulated knowledge and skill (Strober, 1990), and disruptions in social relations (Szreter, 2000). Increase in turnover of employees will result in higher costs due to hiring and training of new employees; and consequently, a lower overall firm performance (Saher, Khan, Shahzad, & Qadri, 2015; Wynen, Van Dooren, Mattijs, & Deschamps, 2019).

Moreover, another major drawback of employees' turnover is the depletion of human capital and skills. Hence, it does not only render a firm inefficient due to limited human capital, but also offer comparative advantage to competitors and new ventures when they absorb such assets (Campbell et al., 2012). Considering the nature of microfinance operation where employees are responsible for marketing microfinance activities, identifying clients, collecting installments, and rendering after service monitoring, social capital[3] is very significant and crucial for sustainability of the industry. In a similar view, Kumar (1999) discovered that long term relationship with clients provided better returns for service-related firms. To further substantiate the argument of social capital, Dess and Shaw (2001) highlighted that when an employee's absence has an influence on an organization internally and externally via various social networks, performance loss will becomes apparent for such firm.

In recent years, majority of the academics opined that the assumptive modeling of linear relationship between turnover and performance may be inappropriate, and consequently suggested a complex and non-linearity relationship instead (Hom, Lee, Shaw, & Hausknecht, 2017; Shaw, 2011; Shaw et al., 2005). The non-linearity assumptions are grounded on the learning curve hypothesis which proposes that the negative effect of turnover decreases with an increase in turnover rates (Lee, 2018). As a result, an inverted U-shaped relationship is promoted by many academics. To support this, De Winne et al. (2019) cited three different effects of small amounts of turnover which include reduction of organizational blindness and groupthink, potential efficiency gains, and high retention cost (Abelson & Baysinger, 1984; Dalton & Todor, 1979; Shaw, 2011). However, after a certain threshold, the cost of turnover outweighs its benefits; and hence, a negative effect on performance becomes evident. This phenomenon is supported by the social and human capital theory, and the comprehensive social network of an organization (De Winne et al., 2019; Lee, 2018) .

[3] Social capital can be defined as "capital and resources which are incorporated in social networks and relationships" (De Winne et al., 2019).

Although employees' turnover has received much attention from the academics, borrower turnover is sparsely discussed in this regard. In a similar vein, we opined that borrower turnover also has a non-linear and inverted U-shaped relationship similar to employees' turnover. Initially, a small amount of borrower (particularly defaulter) turnover might be beneficial for MFIs (Siliki, 2013; Simanowitz, 2000); nonetheless, retaining clients will ultimately reduce administrative costs, lower default risks and increase productivity of MFIs (Pagura, Graham, & Meyer, 2001). However, when the turnover rate rises above the optimal level, attaining new clients becomes costlier than serving the existing ones. In this context, the cost of borrower turnover will outweigh the benefits of attaining new borrowers, and consequently lead to a decline in performance. In this aspects, Pearlman (2014) has rendered several explanations and highlighted the various consequences of borrower dropouts.

3. Methodology

3.1.Modeling Turnover, Gender and Sustainability of MFIs: A Linear Approach

Considering the objective(s) and availability of data, we have decided to utilize the panel data technique to establish the relationship between turnover, gender and sustainability of MFIs. Based on the conventional management literature, we initially assumed a linear relationship between turnover and MFIs sustainability (quadratic relation is reported in next section), and the general empirical expression of the model is as follows;

$$Sus_{i,t} = \beta_0 + \beta_1 Str_{i,t} + \beta_2 Btr_{i.t} + \beta_3 Pfbm_{i,t} + \beta_4 Pfman_{i,t} + \beta_5 Pflo_{i,t} + \beta_6 Pfborr_{i,t} + \beta_7 Dte_{i,t} + \beta_8 Lnsize_{i,t} + \varepsilon_{i,t} \qquad \text{Eq(1)}$$

$$\varepsilon_{i,t} = u_i + \vartheta_{i,t}$$

where sustainability (*Sus*) represent the dimensions of MFIs which (many) have been captured based on the available data and are discussed in the subsequent paragraphs. Moreover, i is the specific MFI, t is the time period, and $\varepsilon_{i,t}$ is the one-way error component model for the disturbances. Hence, u_i is the unobservable individual-specific effect and $\vartheta_{i,t}$ represents purely stochastic disturbance. Many existing studies in the microfinance have utilized the one-way error term in the model (Abdullah & Quayes, 2016; Kyereboah-Coleman, 2007).

The main variable of interest on the left-hand side of the model are staff (*Str*), and borrower turnover rates (*Btr*); representative of gender (*female*) on management, and clients base of MFIs. As discussed earlier, staff turnover is expected to have an effect on various dimensions of MFIs' sustainability. For example, when a loan officer quits, it may be difficult for the new loan officer to collect installments and provide after service monitoring due to unfamiliarity with the borrowers, and localities. However, the problem is acute for firms such as the MFIs where clients' relationship is dispersed among employees, and transfer of such tacit knowledge or relationship to someone else is difficult (Drexler & Schoar, 2014). The worst-case scenario of employees' turnover may result in the reduction of client base as borrowers may lose interest in the firm following the departure of their loan officers; thereby, resulting in lower outreach and a negative impact on financial parameters (Drexler & Schoar, 2014).

Similarly, when a borrower quits or drops out, the long-term relationship between the MFIs and borrowers become severed. MFIs advocate a unique relationship banking for credit services; however, they might experience severe consequences on various sustainability dimensions if a group of disciplined borrowers were to leave the institution who used to attract new members and implement peer mentoring to filter out bad borrowers (Pawlak & Matul, 2004; Urquizo, 2006).

Gender remains another crucial aspect of microfinance since its establishment. Up till date, majority of the MFI clients are females and hence, similar genders tend to maintain a better relationship with their client base; thereby improving the MFIs performance (Agyare, Yuhui, Mensah, Aidoo, & Ansah, 2016; Strøm, D'Espallier, & Mersland, 2014). Furthermore, Thomas and Ramaswamy (1996) also profoundly argued that matching leaders with targeted client groups is an important strategy for the institution to increase its performance. Thus, we have included variables representing females on various management ladders of MFIs as follows: percentage of female board members (*Pfbm*), percentage of female managers (*Pfman*), percentage of female loan officers (*Pflo*), and percentage of female borrowers (*Pfborr*). Based on the earlier studies, it was established that gender dimension has a significant impact on various dimensions of MFIs' sustainability (D'espallier, Guérin, & Mersland, 2011). For example; Hartarska, Nadolnyak, and Mersland (2014) noted that promoting gender diversity at the top management of MFIs can attract financial and social benefits. Moreover, having a significant number of female board members and Chief Executive Officers (CEOs) could improve the outreach base, given the huge population of female clients, and subsequently bolster the financial performance of MFIs (Mersland & Strøm, 2009; Strøm et al., 2014).

Apart from these two sets of variables –turnover and gender, we have equally considered the contribution of leveraging capacity and size of MFIs to sustainability dimensions. The leveraging capacity of MFIs was represented by the debt-to-equity ratio (*Dte*) in this study. The past chunks of literature have highlighted that the capital structure has a significant effect on the performance of MFIs, ranging from financial, to social and environmental factors (Kar, 2012; Khachatryan, Hartarska, & Grigoryan, 2017; Kyereboah-Coleman, 2007; Mia et al., 2018; Tchuigoua et al., 2017). To assess effect of size, various proxies such as the value of plants, machinery, vehicles, equipment, land, and buildings (Cheruiyot, 2017), total assets (Mia & Ben Soltane, 2016), and number of employees (Faruq & David, 2010) have been utilized in the existing studies. However, we have opted for the use of natural logarithm of net fixed asset as a proxy variable of MFIs' sizes to avoid higher correlation with other independent variables (e.g. *Dte*).

The effect of size may present a mixed effect on various sustainability dimensions of MFIs. Among the arguments established for such effects include: first, larger firms are more efficient in managing their resources as they have a diverse pool of human capital, and can utilize economies of scale in operation. Moreover, bigger firms also have a greater market power and easy access to various important resources due to their size effect. Thus a positive association is usually anticipated between the size and performance dimensions. Second, in contrast, a larger firm may also experience coordination problems during a scaling-up operation within the firm, which may lead to lower performance. Third, small firms may be more efficient in using their resources as they are exposed to greater competition with an increased pressure to survive. As a result, they remain up to date in their operation and use cutting edge technology for better financial sustainability.

3.1.1 Turnover, Gender and Cost Sustainability of MFIs

In recent years, MFIs have observed high costs of operation due to intense competition for funding coupled with rapid decline in donations/subsidies; all of which have prompted the practitioners and managers to seek other means of attaining cost sustainability (Islam, Porporato, & Waweru, 2014; Rahman, Mia, Ismail, & Isa, 2018). One of the main reasons commercial banks do not entertain poor clients is due to the high cost and risk associated with processing small scale loans (Pollinger, Outhwaite, & Cordero-Guzmán, 2007). As a result, MFIs utilize social collateral through group lending method to minimize risks and transaction costs. With the continuous increase in employee and borrower's turnover rate, the cost sustainability of MFIs becomes more perplexing. Hence, we investigated the cost sustainability of MFIs by taking two proxy variables, the overall operating expense (*Optexp*), and cost per borrower (*CPB*), and estimated the following two equations (*Eq 1& Eq 2*);

$$Lnoptexp_{i,t} = \beta_0 + \beta_1 Str_{i,t} + \beta_2 Btr_{i.t} + \beta_3 Pfbm_{I,t} + \beta_4 Pfman_{I,t} + \beta_5 Pflo_{I,t} + \beta_6 Pfborr_{I,t} + \beta_7 Dte_{I,t} + \beta_8 Lnsize_{I,t} + \varepsilon_{i,t} \quad \text{Eq(2)}$$

$$Cpb_{i,t} = \beta_0 + \beta_1 Str_{i,t} + \beta_2 Btr_{i.t} + \beta_3 Pfbm_{i,t} + \beta_4 Pfman_{i,t} + \beta_5 Pflo_{i,t} + \beta_6 Pfborr_{i,t} + \beta_7 Dte_{i,t} + \beta_8 Lnsize_{i,t} + \varepsilon_{i,t} \quad \text{Eq(3)}$$

3.1.2. Turnover, Gender and Social Outreach Goal of MFIs.

Since the establishment of modern microfinance in mid 1976, the original aim of the credit movement was to provide services to the poor and financially excluded people. While the objective remains unchanged today, serving the poor has taken several new dimensions, such as breadth (wider service quantity) and depth (serving poorest of the poor-quality) (Schreiner, 2002). To capture these two dimensions, we have utilized the natural logarithm of number of total outstanding loans (*Lnloanout*) to capture the quantity and average loan balance over GNI per capita (*Avlgni*) to represent quality aspects of outreach goals of MFIs based on the existing literature (Mia & Lee, 2017; Mia, Nasrin, & Cheng, 2016; Wijesiri, Yaron, & Meoli, 2017; Yaron, 1994). Considering these two proxy variables, we estimated the following two equations (*Eq 4 & Eq 5*).

$$Lnloanout_{i,t} = \beta_0 + \beta_1 Str_{i,t} + \beta_2 Btr_{i.t} + \beta_3 Pfbm_{I,t} + \beta_4 Pfman_{I,t} \beta_5 Pflo_{i,t} + \beta_6 Pfborr_{i,t} + \beta_7 Dte_{i,t} + \beta_8 Lnsize_{i,t} + \varepsilon_{i,t} \quad \text{Eq(4)}$$

$$Avlgni_{i,t} = \beta_0 + \beta_1 Str_{i,t} + \beta_2 Btr_{i.t} + \beta_3 Pfbm_{i,t} + \beta_4 Pfman_{i,t} + \beta_5 Pflo_{i,t} + \beta_6 Pfborr_{i,t} + \beta_7 Dte_{i,t} + \beta_8 Lnsize_{i,t} + \varepsilon_{i,t} \quad \text{Eq(5)}$$

3.1.3. Turnover, Gender and Financial Sustainability of MFIs.

Along with the outreach goals of MFIs, financial sustainability also carries similar importance in recent years to ensure continuous flow of services to the poor and unbanked (Allet & Hudon, 2015; Lam et al., 2020; Mia et al., 2018). Without standing on their own feet, achieving the poverty alleviation goal of MFIs will be difficult in the long run (Hermes & Lensink, 2007). Unfortunately, it is becoming a reality in the recent years, given the MFIs' adoption of financial system approach mechanism (e.g., microfinance should operate based on the market principles). Nonetheless, financially sustainable MFIs are at advantage to promote entrepreneurship and financial inclusion than their counterpart via spill-over effect (Shankar, 2007). Thus, studies recommend that MFIs should attain certain level of financial sustainability not only to maintain their operation, but also to provide a return to their investors (Hermes &

Lensink, 2007; Lam et al., 2020). Therefore, we represented the possible threat of turnover and gender variables to financial sustainability mission of MFIs by considering two proxy variables – return on assets (*Roa*) and operational self-sufficiency (*Oss*) in the following two equations (*Eq 6 & Eq 7*);

$$Roa_{i,t} = \beta_0 + \beta_1 Str_{i,\,t} + \beta_2 Btr_{i.t} + \beta_3 Pfbm_{i,\,t} + \beta_4 Pfman_{i,\,t} + \beta_5 Pflo_{i,\,t} + +\beta_6 Pfborr_{i,\,t} + \beta_7 Dte_{i,\,t} + \beta_8 Lnsize_{i,\,t} + \varepsilon_{i,t} \quad \text{Eq(6)}$$

$$Oss_{i,t} = \beta_0 + \beta_1 Str_{i,\,t} + \beta_2 Btr_{i.t} + \beta_3 Pfbm_{i,\,t} + \beta_4 Pfman_{i,\,t} + \beta_5 Pflo_{i,\,t} + +\beta_6 Pfborr_{i,\,t} + \beta_7 Dte_{i,\,t} + \beta_8 Lnsize_{i,\,t} + \varepsilon_{i,t} \quad \text{Eq(7)}$$

3.1.2. Turnover, Gender and Institutional Viability of MFIs.

Being an informal lending institution and operating with almost no physical collateral from its clients, the existence of credit risk could pose a credible threat to MFIs' long-term survival (Zamore, Beisland, & Mersland, 2019). To ensure institutional viability, MFIs should minimize its credit risk. Although there is no established factor to represent the survivability of MFIs, we have decided to use two proxy variables (write-off and portfolio-at-risk) related to the credit risk of MFIs, which offer some information on the institution strengths and insolvency. These variables have been used in the existing literature to capture credit risk or loan default of MFIs (Inekwe, 2019). We believe that a higher credit risk will erode the overall market value of MFIs and risky portfolio investment will pose a creditable threat to long term survival. However, it is noteworthy that these two proxies were only employed as proxy measures and not a sole determinant of MFIs survivability. Hence, we estimated the following two equations (*Eq 8 & Eq 9*),

$$Writeoff_{i,t} = \beta_0 + \beta_1 Str_{i,\,t} + \beta_2 Btr_{i.t} + \beta_3 Pfbm_{i,\,t} + \beta_4 Pfman_{i,\,t} + \beta_5 Pflo_{i,\,t} + \beta_6 Pfborr_{i,\,t} + \beta_7 Dte_{i,\,t} + \beta_8 Lnsize_{i,\,t} + \varepsilon_{i,t} \quad \text{Eq(8)}$$

$$Portrisk_{i,t} = \beta_0 + \beta_1 Str_{i,\,t} + \beta_2 Btr_{i,t} + \beta_3 Pfbm_{I,\,t} + \beta_4 Pfman_{i,\,t} + \beta_5 Pflo_{i,\,t} + \beta_6 Pfborr_{i,\,t} + \beta_7 Dte_{i,\,t} + \beta_8 Lnsize_{i,\,t} + \varepsilon_{i,t} \quad \text{Eq(9)}$$

3.2 Modelling Turnover, Gender and Sustainability of MFIs: A Quadratic Approach

Given the non-linear and complex relationship between employees' turnover and organizational performance (De Winne et al., 2019; Hancock et al., 2013; Lee, 2018), we have also assumed the possibility of existence of a non-linear relationship between the two. In addition, the effect of borrowers' turnover may also be beneficial up to a certain level as highlighted in the literature review. Hence, the square terms of the employee and borrower turnovers were added to the original equation and the expanded quadratic expression of *Eq(2)* to *Eq(9)* as follows:

$$LnOptexp_{i,t} = \beta_0 + \beta_1 Str_{i,\,t} + \beta_2 Str_{i,t}^2 + \beta_3 Btr_{i.t} + \beta_4 Btr_{i,t}^2 + \beta_5 Pfbm_{i,\,t} + \beta_6 Pfman_{i,\,t} + \beta_7 Pflo_{i,\,t} + \beta_8 Pfborr_{i,\,t} + \beta_9 Dte_{i,\,t} + \beta_{10} Lnsize_{i,\,t} + \varepsilon_{i,t} \quad \text{Eq(10)}$$

$$Cpb_{i,t} = \beta_0 + \beta_1 Str_{i,\,t} + \beta_2 Str_{i,t}^2 + \beta_3 Btr_{i.t} + \beta_4 Btr_{i,t}^2 + \beta_5 Pfbm_{i,\,t} + \beta_6 Pfman_{i,\,t} + \beta_7 Pflo_{i,\,t} + \beta_8 Pfborr_{i,\,t} + \beta_9 Dte_{i,\,t} + \beta_{10} Lnsize_{i,\,t} + \varepsilon_{i,t} \quad \text{Eq(11)}$$

$$Lnloanout_{i,t} = \beta_0 + \beta_1 Str_{i,t} + \beta_2 Str_{i,t}^2 + \beta_3 Btr_{i.t} + \beta_4 Btr_{i,t}^2 + \beta_5 Pfbm_{i,t} + \beta_6 Pfman_{i,t} + \beta_7 Pflo_{i,t} + \beta_8 Pfborr_{i,t} + \beta_9 Dte_{i,t} + \beta_{10} Lnsize_{i,t} + \varepsilon_{i,t} \quad \text{Eq(12)}$$

$$Avlgni_{i,t} = \beta_0 + \beta_1 Str_{i,t} + \beta_2 Str_{i,t}^2 + \beta_3 Btr_{i.t} + \beta_4 Btr_{i,t}^2 + \beta_5 Pfbm_{i,t} + \beta_6 Pfman_{i,t} + \beta_7 Pflo_{i,t} + \beta_8 Pfborr_{i,t} + \beta_9 Dte_{i,t} + \beta_{10} Lnsize_{i,t} + \varepsilon_{i,t} \quad \text{Eq(13)}$$

$$Roa_{i,t} = \beta_0 + \beta_1 Str_{i,t} + \beta_2 Str_{i,t}^2 + \beta_3 Btr_{i.t} + \beta_4 Btr_{i,t}^2 + \beta_5 Pfbm_{i,t} + \beta_6 Pfman_{i,t} + \beta_7 Pflo_{i,t} + \beta_8 Pfborr_{i,t} + \beta_9 Dte_{i,t} + \beta_{10} Lnsize_{i,t} + \varepsilon_{i,t} \quad \text{Eq(14)}$$

$$Oss_{i,t} = \beta_0 + \beta_1 Str_{i,t} + \beta_2 Str_{i,t}^2 + \beta_3 Btr_{i.t} + \beta_4 Btr_{i,t}^2 + \beta_5 Pfbm_{i,t} + \beta_6 Pfman_{i,t} + \beta_7 Pflo_{i,t} + \beta_8 Pfborr_{i,t} + \beta_9 Dte_{i,t} + \beta_{10} Lnsize_{i,t} + \varepsilon_{i,t} \quad \text{Eq(15)}$$

$$Writeoff_{i,t} = \beta_0 + \beta_1 Str_{i,t} + \beta_2 Str_{i,t}^2 + \beta_3 Btr_{i.t} + \beta_4 Btr_{i,t}^2 + \beta_5 Pfbm_{i,t} + \beta_6 Pfman_{i,t} + \beta_7 Pflo_{i,t} + \beta_8 Pfborr_{i,t} + \beta_9 Dte_{i,t} + \beta_{10} Lnsize_{i,t} + \varepsilon_{i,t} \quad \text{Eq(16)}$$

$$Portrisk90_{i,t} = \beta_0 + \beta_1 Str_{i,t} + \beta_2 Str_{i,t}^2 + \beta_3 Btr_{i.t} + \beta_4 Btr_{i,t}^2 + \beta_5 Pfbm_{i,t} + \beta_6 Pfman_{i,t} + \beta_7 Pflo_{i,t} + \beta_8 Pfborr_{i,t} + \beta_9 Dte_{i,t} + \beta_{10} Lnsize_{i,t} + \varepsilon_{i,t} \quad \text{Eq(17)}$$

The definition of each of the variables used in this study are reported in Table 1.

<Please Insert Table 1 Here>

Since we employed panel data, *Eq (2)* to *Eq (17)* can be estimated via several models, such as pooled ordinary least squares (*POLS*), random effect model (*REM*), and fixed effect model (*FEM*). Nevertheless, the outcome variables in this study that are related to sustainability could also be affected by time-invariant variables, such as legal status, location, regulation, etc. of MFIs. Since we are interested in examining the effect of time variant factors on various sustainability indicators of MFIs, it is suggested to utilize *FEM* rather than the other two models. For example, Abdullah and Quayes (2016) provided explanation as to why *FEM* could be a better choice in the performance investigation of MFIs. A similar fixed effect approach was also employed by Lee (2018) in the study of the US federal agencies to investigate the link between employee's turnover and performance. Moreover, we have estimated the fixed effect model with robust standard errors to overcome any potential heteroskedasticity.

3.3 Data and Sources of Data

In accordance with other past studies (Abdullah & Quayes, 2016; Mia & Ben Soltane, 2016; Quayes, 2015), we have relied on a secondary source of data from the Microfinance Information Exchange (MIX) dataset for this study. Since its establishment in 2002, MIX has been collecting extensive data related to financial service providers across the globe and had gained its reputation in academia due to the quality of its data. Following the self-report of data by respective financial service providers (FSP), MIX then analyzes, standardizes and validates these data in accordance with international standards. Although, the MIX data was freely open for limited variables (and the rest for premium basis), MIX has recently collaborated with the World Bank online catalogue to release its extensive data on MFIs' outreach and financial sustainability variables for free. This was a welcoming initiative by the World Bank and the MIX to promote financial inclusion globally, and serve diverse academic community from the underserved and low-income countries. After extracting the data from 2003 to 2019, our initial screening revealed that prior to 2010 and after 2018, *STR* and *BTR* variables were largely unavailable. This motivated us to utilize data between 2010 and 2018, spanning a 9-year period and containing a total of 2,032 MFIs. Hence, we have employed an unbalanced panel data of 2,032 MFIs from 2010 to 2018 to estimate *Eq (2)* to *Eq (17)*.

4. Results and Discussion

Table 2 present the winsorized (5% and 95% percentile levels) descriptive statistics of the variables used in the study. The report revealed several interesting facts that might present possible implications. For example, the overall cost per borrower was US$204.084, while the minimum and maximum cost per borrower was US$12 and US$ 694 respectively. The higher cost per borrower may be attributed to the overall loan size, geography, and delivery mechanism. Interestingly, the average loan size for the sample mean was around half of the Gross National Income (GNI) and could increase up to 2.5 times the value. Based on the mean *Roa*, it was discovered that many MFIs recorded positive returns; although, very small (around 1.8%), while one or more MFIs reported a negative return. In general, the mean value of operational self-sufficiency was above 1 which implies that the sector was operationally sustainable as a whole. Interestingly, the mean value of the write-off ratio is almost equal to *Roa*, indicating that MFIs have some risky investments in their portfolio. Considering the portfolio at risk for a 90-day period, the rate is alarming, given the mean value which is just above 4 %.

Most importantly, the average staff's turnover rate stood at 20%, which is relatively high and could increase up to 65%. Despite this, some MFIs still did not observe any staff leaving the organization, given the minimum value of 0. Similarly, the mean value of borrowers' turnover is staggering with a value of 23.6%, which is higher than the staff's turnover. In terms of gender representation in the management of MFIs, we observed that female loan officers constitute around 39% of the total labor force, while a little over one-third make up the managerial posts. Interestingly, around 30% of MFIs board members were female. Although, there exist at least one or more MFIs with no females in their management team, our sample also include MFIs that are totally run by female managers and loan officers. In terms of borrowers, females constitute the largest client base of MFIs with an average value of 65%. Besides, there were no MFIs in our sample without female borrowers; consequently, this reiterates the original objective of microfinance as being a female-centered development initiative. In terms of the leveraging capacity of MFIs, debt amounting to just over three and half times the value of the equity was observed on average. Due to its long existence, MFIs may prefer to rely on debt financing (Kyereboah-Coleman, 2007). Moreover, there is also a minimum of one MFI in the sample that have used more equity than debt given the minimum value of less than 1.

<Please Insert Table 2 here>

Multicollinearity remains an important challenge in regression analysis, and therefore requires thorough observation. After estimating pairwise correlation and variance inflation factors (VIF); the findings revealed the absence of multicollinearity, as the value remains well below the conventionally acceptable limit (usually 0.8 and 10 for pairwise correlation and VIF respectively)(Kennedy, 2008; O'brien, 2007). Thus, we can conclude that the empirical model estimated below does not experience a setback from a multicollinearity.

<Please Insert Table 3 Here>

4.1 Results of the Linear Regression

At the onset, only the linear regression was considered, and *Eq(2)* to *Eq(9)* were estimated using fixed effect model with robust standard errors. The results were reported in Tables 4, 5, 6 and 7 for cost sustainability, social outreach goals, financial sustainability and survivability of MFIs respectively. The overall fitness of the models were good as F-statistics remains significant for all models (except for model 2). Moreover, the explanatory power measured by R^2 was relatively low for some models but can still be accepted. The results of the main variables (turnover and gender) were initially discussed before the control variables. Also, caution should be exercised when interpreting the findings as the sign of the coefficients may have different implications for various sustainability dimensions of MFIs (see Table 1).

As indicated in Table 4, borrower turnover was observed to be negatively related to cost sustainability and remains significant (except. Model 3) for both proxies. In contrast, staff turnover was discovered to be positive and significant at the 5% level only for model 4. However, the *Str* variable loses its significance upon introduction of the gender and control variables in the model. Based on the aforementioned, we can conclude that the borrower turnover will reduce the operational costs of MFIs. Several plausible explanations were offered for such outcomes; for example, MFIs may intentionally drop borrowers that are costly to maintain from their client-base, and consequently improve the institution's efficiency. In the same vein, serving female borrowers was discovered to be cost-effective given the negative and statistically significant effect on cost per borrower. Unfortunately, serving female borrowers has no significant effect on the overall operating expenses.

<Please Insert table 4 Here>

Surprisingly, staff turnover was found to have no significant effect on the social outreach goal of MFIs for both proxies used. In contrast, the result revealed that borrower turnover leads to the reduction in MFIs' loan disbursement and preferential service to relatively wealthier clients (higher *Avlgni* implies relatively wealthier clients). In other words, the social outreach goal of MFIs will be compromised as borrower's turnover rate increases. It could be inferred from the high *Avlgni* that MFIs may need to increase the average loan size, so as to maintain the targeted financial return and overall size of the loan portfolio. Interestingly, our results also revealed that female managers promote financial inclusion and depth of outreach as observed in the statistically significant positive and negative coefficient of the number of loan outstanding and average loan size respectively. Our findings also reiterated that female borrowers usually demand smaller loan size, as it has a negative and statistically significant relationship with *Avlgni*.

<Please Insert Table 5 Here>

The effect of turnover on financial sustainability is very alarming as depicted in the negative effects of *Str* and *Btr* on two proxies used to measure financial sustainability which are statistically significant at 1% levels for all models (Models - 13 to 18). Recently, MFIs struggle to attain financial sustainability, and the negative consequences of turnover rates makes it even worse. The findings support the conventional claim that employees' turnover will have a negative consequence on the financial sustainability of an organization (Park & Shaw, 2013).

In terms of gender, female managers might be better for MFIs due to their positive and statistically significant effect on the financial sustainability of MFIs. Although it was argued that female loan officers had better capacity to build trust with the clients and enforce monitoring activities (van den Berg, Lensink, & Servin, 2015), our findings indicated otherwise, except for the argument that female employees perform better in managerial roles than field force. Furthermore, Khachatryan et al. (2017) reported a statistically significant effect of female borrowers on *Roa* of MFIs, which is in contrast with our observation of a positive but negligible effect. Besides, our results disagree with that of Bogan (2012) on the relationship between size and *Roa*.

<Please Insert Table 6 Here>

Based on the result presented in Table 7, we concluded that turnover increases the credit risk of MFIs and may jeopardize their long-term survivability. The results also supported our arguments presented earlier that departure of staff hampers the timely collection of installments, resulting in the rise of write-off ratio and portfolio at risk at 90 days. More so, a significant amount of loan will also be written-off with an increase in borrower turnover rates as indicated in the positive coefficient sign which is statistically significant at 1% levels across the models (Model-19 to 24). Since turnover has a positive and significant relationship with write-off and Portrisk90, it implies that MFI might struggle to survive in the long run if the turnover rates continue to rise. This finding reiterates the result presented in Table 6. Interestingly, female borrowers were discovered to have a negative effect on the Portrisk90, implying they were disciplined borrowers and repaid installments on time.

<Please Insert Table 7 Here>

In terms of the leveraging capacity of MFIs, we discovered a contrasting effect on various MFIs sustainability dimensions. For example, increase in *Dte* results in significantly higher overall operating expenses and cost per borrower. In addition, more debt financing ultimately increases the number of loan outstanding, which is associated with a better breadth of outreach performance of MFIs. This result concurs with the findings of Kyereboah-Coleman (2007) but contradicts Kar (2012). Also, more debt financing affects the financial sustainability proxies (both *Roa* & *Oss*) of MFIs, given the negative effect with statistical significance. Additionally, increase in the leveraging capacity of MFIs (↑ *Dte*) results in a corresponding increase in the portfolio risk.

As hypothesized earlier, size was found to exhibit an irreconcilable effect on various sustainability dimensions of MFIs. As an instance, in terms of cost sustainability, large MFIs incur significantly high operating costs, as well as cost per borrower. On the other hand, in terms of outreach sustainability, larger MFIs promote breadth of outreach (Model-9), and simultaneously depress the depth of outreach by lending to relatively wealthier clients (model-12). However, no statistically significant effect of size was observed on financial and survivability dimensions of MFIs.

4.2 Results of Non-linear (Quadratic) Regression

The results of the quadratic *Eq(10)* to *Eq(17)* are reported in Table 8. After including quadratic terms in the equation, significance levels of *Str* and *Btr* was observed to have changed compared with the linear models in many proxies of MFIs sustainability. Nevertheless, there is an 'inverted' U-shaped relationship between *Str* and all proxies of sustainability dimensions used in the study (though they are not significant). However, the observed inverted U-shaped relationship is weak and statistically insignificant except for *Str* and Str^2 in Model-25. This implies that increasing the value of *Str* will raise the overall cost of operation up to a certain threshold, after which further increase will result in lower operating costs. This observation is best explained by the fact that when the employees' turnover is low, MFIs incur more cost due to training and supervision of a new employee. However, when *Str* is relatively high, MFIs spend less as new employees bring some innovative ideas, techniques and innovations that will ultimately result in overall cost reduction. The claim is further substantiated by the inverted U-shaped relationship of *Cbp*. Similarly, it was discovered that increasing *Str* to a certain extent poses no harm until after an optimum level, where it begins to affect the financial sustainability dimensions of MFIs. Furthermore, the quadratic equation also reveals that *Str* will result in a higher write-off ratio; however, we did not observe any significant effect of *Str* on outreach dimensions of MFIs even after considering the quadratic regression.

The inclusion of Btr^2 also revealed that the relationship with various dimensions of MFIs sustainability could be U-shaped/inverted, or linear. For example, borrower turnover has an inverted U-shaped relationship with cost sustainability, financial sustainability and breadth of outreach; a linear relationship with the depth of social outreach and survivability (Portrisk90 only); and a U-shaped relationship with survivability (Write-off only) dimensions of MFIs. Moreover, the negative and significant coefficient sign of Btr^2 may highlight the risk of MFIs halting operations in a particular branch or area that was deemed too costly to maintain (Model 25 & 26). The inverted U-shaped relationship with financial sustainability may reflect that alleged exclusion of bad borrowers could enhance overall financial sustainability (although it is insignificant); however, negative consequences may ensue when the turnover rates exceeds a certain threshold. This observation is backed by the negative effect of *Btr* on write-off ratio.

Apart from that, the effect of gender, leveraging capacity and size variables on various sustainability dimensions of MFIs in the quadratic model remain almost the same, which signifies the robustness of the result.

<Please Insert Table 8 Here>

Despite the weak quadratic relationship with outcome variables, we are still interested to see the vertex point, where the parabola changes its direction. With this knowledge, concise estimates will be provided to the managers of MFIs to enable them to understand when the turnover (employee and borrower) will have a positive or negative effect on various dimensions of MFIs sustainability. In this section, we only discussed variables that were significant in at least one of the models (linear or non-linear coefficient) or both.

We discovered that the vertex point of *Str* to operating expense was 0.366 or 36.6%, after which the direction of parabola changes its sign with an inverted U-shaped relationship of statistical significance. In other words, up to the 36.6% level, the operating costs will increase and start to reduce afterwards. Similarly, up to 6.3% of *Str*, *Roa* exhibits a positive effect on financial sustainability, after which a further increase will have significant negative effect. In terms of operational self-sufficiency (*Oss*), the vertex point is at 17.1% which implies that the employees' turnover up to the 17.1% mark will be positive to *Oss*, after which it becomes negative and statistically significant.

<Please Insert table 9 Here>

In terms of borrower turnover, we discovered that up to 16.5% and 14.4% of operating expenses and *Cpb* respectively, MFIs will observe increase in overall costs and cost per borrower (although they are insignificant). However, if the borrower turnover rates exceed these vertex points, it is statistically found to have a cost reduction effect on MFIs subsequently. Moreover, our results also revealed that when borrower turnover exceeds 10.3%, the number of loan outstanding of an MFIs also reduces, thereby resulting in a negative effect on the breadth of social outreach. The vertex point of *Btr* in relation to *Roa* also reiterates that borrower turnover is detrimental to MFIs financial sustainability. MFIs will observe significant negative effect on *Roa* and *Oss* when the borrower turnover rate exceeds 7.5% and 3.7% respectively. As expected, borrower turnover rates above 7.6% will result in a higher write-off ratio and threaten the survivability of MFIs. All these vertex points are within the data range, hence the findings are credible.

5. Conclusion, Policy Implications and Future Research

By utilizing a recent and large scale globally represented panel data set, the study has evaluated the effect of employee and borrower's turnover on various sustainability dimensions of MFIs. In line with the existing literature, we have exploited both the linear and quadratic relationship, thereby producing laudable results. Our results strongly support the evidence of linear relationship of turnover with sustainability of MFIs. Having said that, we also discovered the presence of an inverted U-shaped (weak) relationship between employee's turnover and sustainability. Similarly, for borrower's turnover, the relationship can be linear/U-shaped, or inverted/U-shaped; however, linearity is strongly supported in most of the models. Also, inclusion of gender variables reveals that women in all organizational ladders are not beneficial for MFIs' sustainability as we have found mix/insignificant results. The effect of leveraging capacity of MFIs and size were found to have a conflicting effect depending on the aspects of sustainability.

Since we have observed that employee's turnover has a negative effect on most of the dimensions of MFIs sustainability, the management of MFIs should emphasize on internal and external marketing approaches as suggested by Kanyurhi Eddy and Bugandwa Mungu Akonkwa (2016). In a similar vein, Woller (2002) proposed promotion of internal and external marketing practices within MFIs so as to boost the staff morale and enhance their productivity, and ultimately overcome staff turnover. This is the reason for the proposal of the staff incentive scheme by Pityn and Helmuth (2005); and Holtmann and Grammling (2005) observed that such incentive scheme has a credible effect on reducing staff turnover and enhancing productivity.

Moreover, Kanyurhi Eddy and Bugandwa Mungu Akonkwa (2016) have further highlighted the importance of fair reward and more autonomy to staff. By giving more power and autonomy to the decision-making process, employees will be more dedicated and satisfied as a result of the feeling of self-worth perceived within the firm. If employees are not heard, valued and treated fairly by the top management, they will become stressed, frustrated and consequently leave the organization (Laddha, Singh, Gabbad, & Gidwani, 2012). Based on the social exchange theory, Herman, Huang, and Lam (2013) proposed that effective leadership training program in lieu to promote employee's emotional attachment with the firm could minimize employee's turnover. To some extent, forced borrower dropout can be beneficial provided MFIs can filter bad borrowers who do not pay instalments on time or defaulters. However, to avoid huge borrower turnover; MFIs should undertake various retention policies, such as product diversification and incentives.

In terms of gender variables in the ladder of MFIs management, we discovered that female managers contribute more to sustainability of MFIs. Hence, integrating more female into managerial roles would be beneficial for MFIs. In contrast, MFIs should reconsider the employment of female loan officers, as it was discovered to be detrimental to the MFIs' sustainability. Although existing studies have highlighted the importance of female board members on sustainability, we failed to observe any significant effect of this variable on all the models. Also, MFIs should be cautious in utilizing debt financing in their operations and perform cost-benefit analysis before making any decision. Similarly, the relative high cost of serving borrowers in large scale MFIs could be attributed to the inefficiency in using resources and coordination problems, consequently warranting the attention of the management.

Despite a thorough examination of effects of turnover and gender on various sustainability dimensions of MFIs, we acknowledged that the study suffers some limitations, one of which is the relation of estimated models to the explanatory power. Hence, future studies may include more contextual variables and consider dynamic modelling techniques for the possibility of enhancing the explanatory power. Second, borrower and staff's turnover can be represented in many forms not included in this study. For example, a borrower may graduate from a microfinance program and voluntarily leave or dropout due to default. The effect of these types of borrower turnover on sustainability dimensions of MFIs may differ, which we could not capture due to unavailability of data. Similarly, we lacked access to specific data on various types of staff turnover rates, such as voluntary, forced, or transfer. Thus, we could not observe the effect of these specific turnover rates on sustainability, as our study is limited to studying the aggregate effect. Also, future consideration should be given to the decomposed data on turnover rates as it could provide more concise estimates of the parameters. In conclusion, we only explored the linear and quadratic relationship between turnover and sustainability, but the relationship might be more complex than observed, given the limitation of this study (De Winne et al., 2019). Therefore, future studies should consider exploring more complex and flexible curvilinear relationship between turnover and various performance dimensions of MFIs.

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Table 1: Definition of Variables.

Variable	Definition	Remark
Operating expense (Lnoptexp)	Includes expenses not related to financial and credit loss impairment, such as personnel expenses, depreciation, amortization and administrative expenses.	Lower is better
Cost per borrower (Cpb)	Operating Expense / Average Number of Active Borrowers	Lower is better
Number of loans outstanding (Lnnloan)	Natural logarithm of the number of loans in the gross loan portfolio.	Higher is better
Avlgni	Average Loan Balance per Borrower / GNI per Capita	Lower is better
Return on Assets (Roa)	(Net Operating Income - Taxes) / Average Total Assets	Higher is better
Operational self-sufficiency (Oss)	Financial Revenue / (Financial Expense + Net Impairment Loss + Operating Expense)	Higher is better
Write-off ratio	Value of loans written-off / Average Gross Loan Portfolio	Lower is better
Portfolio at risk>90 days (Portrisk90)	Outstanding balance, portfolio overdue > 90 Days + renegotiated portfolio / Gross Loan Portfolio	Lower is better
Staff turnover rate (Str)	Percentage of staff (permanent and contract) having left the financial institution during the last reporting year divided by the average number of permanent and contract staff for the period	-
Staff turnover rate square (Str^2)	Str*Str	+/-
Borrower turnover rate (Btr)	1- borrower retention rate (BRR)[a]	-
Borrower turnover rate square (Btr^2)	Btr*Btr	+/-
Pfbm	Percent of female board members	+
Pfman	Percent of female loan officers	+
Pflo	Percent of female managers	+
Pfborr	Number of active female borrowers / Number of Active Borrowers	+
Debt to equity ratio (Dte)	Total Liabilities / Total Equity	+/-
Lnsize	Natural logarithm of total assets measured in USD$	+/-

Source: Authors compilation from the World Bank field definition. [a]BRR=Active borrowers at the end of the reporting period divided by the sum of active borrowers at the beginning of the reporting period and new borrowers during the reporting period.

Table 2: Descriptive statistics (full sample)

Variable	Obs.	Mean	Std. Dev.	Min	Max
Lnoptexp	7492	14.175	1.778	10.745	17.056
Cpb	6691	204.084	194.152	12.000	694.000
Lnloan	8184	9.379	1.938	5.740	12.786
Avglgni	7935	0.545	0.656	0.040	2.460
Roa	6881	0.018	0.054	-0.140	0.110
Oss	7475	1.142	0.266	0.580	1.740
Writeoff	6270	0.019	0.026	0.000	0.090
Portrisk	7067	0.041	0.049	0.000	0.190
Str	5412	0.207	0.172	0.000	0.650
Btr	4226	0.236	0.142	0.000	0.530
Pfbm	5837	0.304	0.235	0.000	0.860
Pfman	6251	0.354	0.299	0.000	1.000
Pflo	6421	0.387	0.291	0.000	1.000
Pfborr	7014	0.653	0.262	0.210	1.000
Dte	7706	3.631	3.217	0.040	12.500
Lnsize	7635	12.471	2.019	8.559	15.802

Source: Authors estimate based on the World Bank data. Winsorized all variables at 5% levels. Gender variables were not multiplied by 100 similar with the original dataset.

Table 3: Variance Inflation Factors and pairwise correlation between independent variables.

	VIF	Str	Btr	Pfbm	Pfman	Pflo	Pfborr	Dte	Lnsize
Str	1.03	1.000							
Btr	1.03	0.125	1.000						
Pfbm	1.13	-0.078	0.011	1.000					
Pfman	1.4	-0.038	-0.016	0.320	1.000				
Pflo	1.36	-0.043	0.006	0.270	0.481	1.000			
Pfborr	1.09	0.115	-0.059	0.127	-0.003	0.087	1.000		
Dte	1.09	-0.033	-0.062	-0.065	-0.134	-0.098	-0.006	1.000	
Lnsize	1.13	-0.023	-0.061	-0.118	-0.171	-0.154	-0.187	0.246	1.000
Mean VIF	1.16								

Source: Authors estimate based on the World Bank data.

Table 4: Effect of turnover on cost sustainability of MFIs (Fixed Effect Model).

	Dependent Variable: LnOptexp			Dependent Variable: Cpb		
	Model-1	Model-2	Model-3	Model-4	Model-5	Model-6
Str	0.100	0.130	0.087	24.670**	17.956	17.456
	(0.095)	(0.112)	(0.095)	(11.034)	(11.710)	(11.464)
Btr	-0.243**	-0.218**	-0.150	-37.666***	-33.172**	-30.322**
	(0.099)	(0.110)	(0.099)	(12.642)	(14.079)	(14.126)
Pfbm		0.008	-0.053		23.754	19.960
		(0.114)	(0.096)		(15.225)	(15.133)
Pfman		0.127	0.144*		-24.153*	-23.104
		(0.079)	(0.074)		(14.517)	(14.485)
Pflo		-0.168	-0.164		-10.646	-8.215
		(0.139)	(0.118)		(22.418)	(21.791)
Pfborr		-0.192	-0.112		-59.106***	-55.707**
		(0.212)	(0.195)		(22.618)	(23.052)
Dte			0.021**			-1.518
			(0.009)			(0.965)
Lnsize			0.373***			17.881***
			(0.036)			(5.028)
Cons	14.619***	14.773***	9.836***	220.045***	260.837***	33.331
	(0.029)	(0.157)	(0.493)	(3.585)	(17.055)	(67.444)
Observations	3512	2999	2977	3512	2999	2977
F statistics	3.362**	1.605	15.961***	6.426***	3.430***	4.800***
R^2	0.004	0.007	0.278	0.006	0.014	0.050
Adj. R^2	0.003	0.005	0.276	0.006	0.012	0.048
# of MFIs	1151	1033	1029	1151	1033	1029

Source: Authors estimate based on the World Bank data. Standard (robust) errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Effect of turnover on social outreach of MFIs (Fixed Effect Model).

	Dependent Variable: Lnnloan			Dependent Variable: Avlgni		
	Model-7	Model-8	Model-9	Model-10	Model-11	Model-12
Str	0.020	0.061	0.039	-0.004	-0.001	-0.007
	(0.082)	(0.093)	(0.085)	(0.035)	(0.034)	(0.035)
Btr	-0.629***	-0.579***	-0.567***	0.115***	0.127***	0.152***
	(0.089)	(0.091)	(0.087)	(0.041)	(0.044)	(0.044)
Pfbm		-0.003	-0.031		0.020	0.009
		(0.103)	(0.093)		(0.054)	(0.054)
Pfman		0.206***	0.218***		-0.074*	-0.079*
		(0.068)	(0.068)		(0.040)	(0.041)
Pflo		-0.043	-0.065		-0.105	-0.089
		(0.107)	(0.106)		(0.064)	(0.060)
Pfborr		0.101	0.179		-0.430***	-0.386***
		(0.171)	(0.160)		(0.145)	(0.147)
Dte			0.030***			0.005
			(0.008)			(0.003)
Lnsize			0.217***			0.063***
			(0.025)			(0.019)
Cons	9.917***	9.835***	6.904***	0.514***	0.851***	-0.019
	(0.025)	(0.132)	(0.354)	(0.012)	(0.100)	(0.258)
Observations	3627	3089	2984	3536	3010	2912
F statistics	25.358***	8.520***	20.276***	4.104**	2.987***	3.428***
R^2	0.035	0.037	0.194	0.004	0.032	0.073
Adj. R^2	0.034	0.035	0.192	0.003	0.030	0.070
# of MFIs	1205	1070	1029	1201	1065	1027

Source: Authors estimate based on the World Bank data. Standard (robust) errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Effect of turnover on financial sustainability of MFIs (Fixed Effect Model).

	Dependent Variable: Roa			Dependent Variable: Oss		
	Model-13	Model-14	Model-15	Model-16	Model-17	Model-18
Str	-0.029***	-0.029***	-0.028***	-0.110***	-0.107***	-0.102***
	(0.008)	(0.009)	(0.009)	(0.034)	(0.036)	(0.037)
Btr	-0.030***	-0.033***	-0.033***	-0.152***	-0.172***	-0.170***
	(0.008)	(0.009)	(0.008)	(0.032)	(0.036)	(0.036)
Pfbm		0.006	0.006		0.006	0.004
		(0.008)	(0.008)		(0.037)	(0.037)
Pfman		0.012*	0.013**		0.070**	0.073**
		(0.006)	(0.006)		(0.029)	(0.029)
Pflo		-0.026**	-0.026**		-0.090*	-0.089*
		(0.011)	(0.011)		(0.046)	(0.046)
Pfborr		0.008	0.007		0.056	0.048
		(0.013)	(0.013)		(0.063)	(0.066)
Dte			-0.002***			-0.011***
			(0.001)			(0.003)
Lnsize			-0.002			-0.010
			(0.001)			(0.008)
Cons	0.034***	0.033***	0.064***	1.214***	1.186***	1.364***
	(0.003)	(0.009)	(0.020)	(0.011)	(0.046)	(0.106)
Observations	3478	2975	2961	3505	2993	2970
F statistics	13.147***	6.130***	7.015***	15.251***	6.577***	7.268***
R^2	0.020	0.029	0.046	0.019	0.027	0.042
Adj. R^2	0.019	0.027	0.044	0.018	0.025	0.039
# of MFIs	1144	1026	1023	1150	1032	1028

Source: Authors estimate based on the World Bank data. Standard (robust) errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Effect of turnover on survivability of MFIs (Fixed Effect Model).

	Dependent Variable: Writeoff			Dependent Variable: Portrisk90		
	Model-19	Model-20	Model-21	Model-22	Model-23	Model-24
Str	0.011^{***}	0.014^{***}	0.013^{***}	0.025^{***}	0.024^{***}	0.021^{***}
	(0.003)	(0.004)	(0.004)	(0.008)	(0.009)	(0.008)
Btr	0.029^{***}	0.029^{***}	0.030^{***}	0.035^{***}	0.034^{***}	0.036^{***}
	(0.004)	(0.005)	(0.005)	(0.008)	(0.009)	(0.008)
Pfbm		-0.003	-0.003		-0.013	-0.013
		(0.003)	(0.003)		(0.008)	(0.008)
Pfman		-0.003	-0.004		-0.003	-0.004
		(0.003)	(0.003)		(0.006)	(0.006)
Pflo		0.002	0.002		0.011	0.013
		(0.004)	(0.004)		(0.008)	(0.008)
Pfborr		0.002	0.004		-0.031^{**}	-0.029^{*}
		(0.007)	(0.007)		(0.015)	(0.015)
Dte			0.000			0.002^{***}
			(0.000)			(0.001)
Lnsize			0.000			0.000
			(0.001)			(0.002)
Cons	0.008^{***}	0.007	0.000	0.026^{***}	0.047^{***}	0.031
	(0.001)	(0.006)	(0.010)	(0.003)	(0.011)	(0.022)
Observations	3392	2894	2806	3501	3002	2906
F statistics	31.953^{***}	8.813^{***}	6.957^{***}	12.903^{***}	4.665^{***}	4.795^{***}
R^2	0.044	0.048	0.051	0.025	0.030	0.046
Adj. R^2	0.043	0.046	0.049	0.024	0.028	0.043
# of MFIs	1162	1031	994	1169	1047	1006

Source: Authors estimate based on the World Bank data. Standard (robust) errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Quadratic relationship between turnover and sustainability of MFIs (FEM).

Dependent Variable:	Lnoptexp	Cpb	Lnnloan	Avglgni	Roa	Oss	Writeoff	Portrisk
	Model-25	Model-26	Model-27	Model-28	Model-29	Model-30	Model-31	Model-32
Str	0.581**	75.806*	0.273	0.150	0.007	0.125	0.022*	0.036
	(0.276)	(42.424)	(0.255)	(0.133)	(0.022)	(0.101)	(0.012)	(0.024)
Str^2	-0.794*	-93.226	-0.356	-0.260	-0.056*	-0.365**	-0.015	-0.025
	(0.414)	(63.399)	(0.398)	(0.195)	(0.034)	(0.152)	(0.019)	(0.037)
Btr	0.321	43.346	0.409	0.083	0.015	0.031	-0.013	0.015
	(0.316)	(46.121)	(0.265)	(0.143)	(0.025)	(0.108)	(0.013)	(0.022)
Btr^2	-0.971*	-150.879*	-1.976***	0.132	-0.100*	-0.415*	0.085***	0.041
	(0.563)	(89.493)	(0.514)	(0.311)	(0.054)	(0.227)	(0.030)	(0.049)
Pfbm	-0.055	19.705	-0.030	0.007	0.006	0.003	-0.003	-0.013
	(0.096)	(14.949)	(0.093)	(0.054)	(0.008)	(0.037)	(0.003)	(0.008)
Pfman	0.139*	-23.967*	0.206***	-0.078*	0.013**	0.071**	-0.003	-0.004
	(0.074)	(14.462)	(0.067)	(0.041)	(0.006)	(0.028)	(0.003)	(0.006)
Pflo	-0.162	-7.678	-0.052	-0.092	-0.025**	-0.089*	0.002	0.013
	(0.117)	(21.666)	(0.104)	(0.060)	(0.011)	(0.047)	(0.004)	(0.008)
Pfborr	-0.101	-53.955**	0.201	-0.389***	0.008	0.053	0.003	-0.029*
	(0.195)	(23.087)	(0.158)	(0.147)	(0.013)	(0.065)	(0.007)	(0.015)
Dte	0.020**	-1.630*	0.029***	0.005	-0.003***	-0.011***	0.000	0.002***
	(0.009)	(0.966)	(0.008)	(0.003)	(0.001)	(0.003)	(0.000)	(0.001)
Lnsize	0.372***	17.730***	0.216***	0.063***	-0.002	-0.011	0.000	0.000
	(0.036)	(5.013)	(0.025)	(0.019)	(0.001)	(0.007)	(0.001)	(0.002)
Cons	9.761***	22.861	6.803***	-0.021	0.057***	1.331***	0.003	0.032
	(0.491)	(67.634)	(0.354)	(0.259)	(0.020)	(0.105)	(0.010)	(0.022)
N	2977	2977	2984	2912	2961	2970	2806	2906
F	14.673***	4.129***	17.486***	2.913***	5.853***	6.606***	5.692***	4.060***
r2	0.281	0.054	0.204	0.074	0.051	0.047	0.060	0.047
r2_a	0.279	0.051	0.202	0.071	0.047	0.044	0.057	0.044
# of MFIs	1029	1029	1029	1027	1023	1028	994	1006
Utest(employee turnover)	1.54*	1.06	0.65	1.13	0.30	1.24	[a]	[a]
Utest(Borrower turnover)	1.02	0.94	1.55*	[a]	0.62	0.29	0.96	[a]

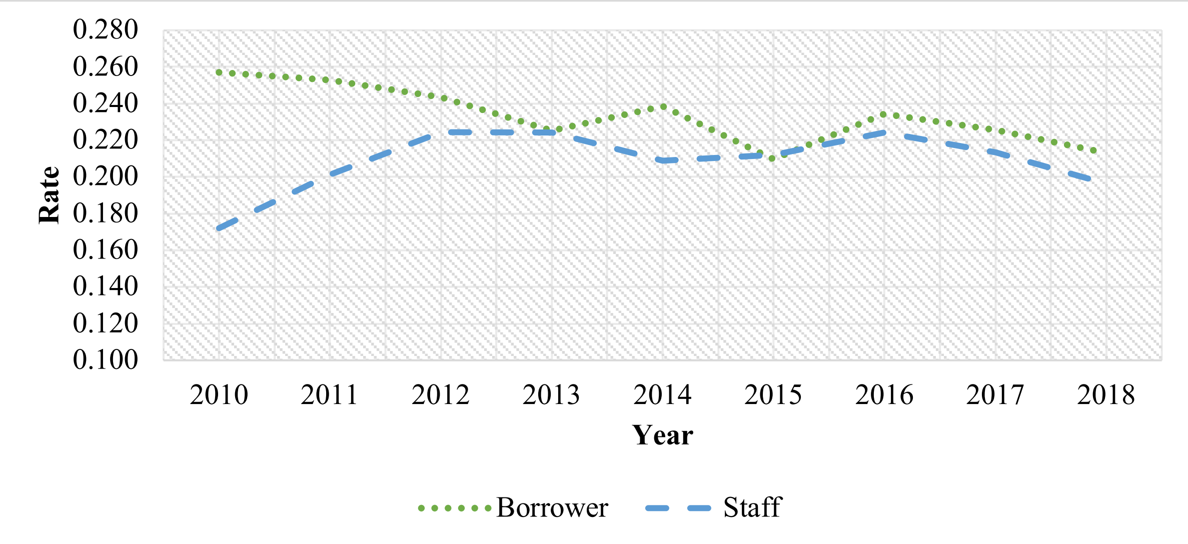
Source: Authors estimate based on the World Bank data. Standard (robust) errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. We have executed Lind–Mehlum (*utest*) command to test for a U-shaped relationship. *utest* report *t* value. [a] Extremum outside interval - trivial failure to reject H_0 (H_0: monotone or U shape).

Table 9: Vertex point of quadratic equation.

Variable	Vertex point for employee turnover	Vertex point for borrower turnover
Lnoptexp	0.366	0.165
Cpb	0.407	0.144
Lnnloanout	0.383	0.103
Avglgni	0.288	-0.314
Roa	0.063	0.075
Oss	0.171	0.037
Writeoff	0.733	0.076
Portrisk	0.720	-0.183

Source: Author. Note: Vertex point = -linear coefficient/ (2* quadratic coefficient).

Figure 1: Staff and borrower turnover rates in the global microfinance industry (2010-2018).



Source: Author based on the World Bank data.

Note: We have used industry mean value both for Str and Btr. Number of observations (within bracket) used to calculate mean value and draw Figure 1 is as follows:
Borrower turnover rate: 2010(675) 2011(550) 2012(505) 2013(377) 2014(420) 2015(484) 2016(474) 2017(402) 2018(339).
Staff turnover rate: 2010(796) 2011(778) 2012(684) 2013(543) 2014(590) 2015(638) 2016(526) 2017(458) 2018(399)